

ECE330 – Spring 2024
Exam 1
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90 min

Thursday, February 22, 2024

NAME _____

NetID _____

Instructions:

- Clearly PRINT your name in CAPITAL LETTERS.
- Clearly write your NetID.
- This exam is closed book and closed notes.
- Please SHOW ALL calculations and steps to receive full credit.
 - Writing just the correct numerical answer with no derivation will not result in credit.
 - Writing just the formula will not result in partial credit.
- You may write on both sides of the exam paper. If there is anything important on the reverse side, please indicate this on the front side.
- All answers must INCLUDE UNITS whenever appropriate.
- Graphing and/or 4-function calculators are permitted, but no cellphones/laptops.
- This exam booklet contains 9 pages (including this one) with 4 problems.

Problem 1 (25 Points)

A single-phase source serves two loads connected in parallel.

- Load 1 draws 180 W at a lagging PF of 0.6.
- Load 2 draws 1 A at a leading PF of 0.6.

The voltage across the loads is $\bar{V} = 100\angle 30^\circ$ V. (Note that the voltage angle is not zero.)

a) Compute the current phasor $\bar{I}_1$ for the first load (180 W at a lagging PF of 0.6). Your answer should be a complex number, in either rectangular or polar form, whichever is more convenient.

$\bar{I}_1 =$ _____ Unit= _____

b) Compute the current phasor $\bar{I}_2$ for the second load (1 A at a leading PF of 0.6). Your answer should be a complex number, in either rectangular or polar form, whichever is more convenient.

$\bar{I}_2 =$ _____ Unit= _____

c) How many VARs of capacitance do we need to make the overall PF unity? Your answer should be stated as a numerical value.

Capacitance= _____ Unit= VAR

d) How many VARs of capacitance do we need to make the overall PF 0.8 leading? Your answer should be stated as a numerical value. Please indicate your reasoning for why your answer should correctly yield a leading and not lagging overall PF.

Capacitance= _____ Unit= VAR

Reason why answer above gives overall leading (and not lagging)

Problem 2 (25 Points)

A 3-phase, 208 V (line-line), Wye-connected source is supplying two 3-phase loads connected in parallel:

1. Wye-connected, inductive, draws 10 A phase current and 3100 W of real power (3-phase)
2. Delta-connected, it has per-phase impedance $\bar{Z}_2$, draws 8 A line current and 1800 VARs of reactive power (3-phase) at a leading power factor.

a) Find the total real power (3-phase) supplied by the source.

$P_{\text{tot}} =$ _____ Unit= _____

b) Compute the total line current (magnitude) being supplied by the source.

$I_L =$ _____ Unit= _____

c) Find per-phase impedance of the second load, $\bar{Z}_2$

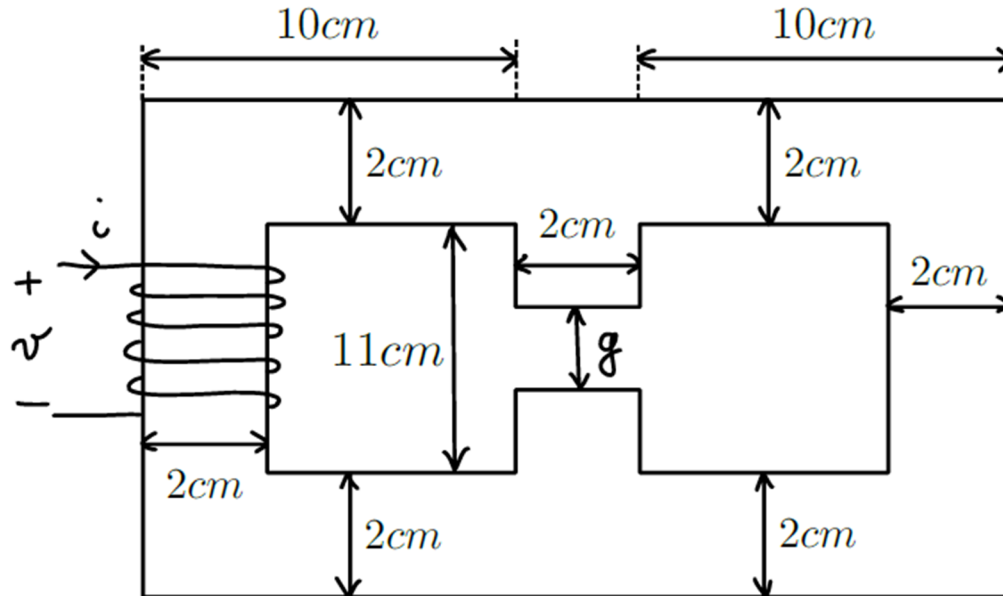
$\bar{Z}_2 =$ _____ Unit= _____

d) Now we reconfigure the second load to be connected in Wye (instead of Delta, as originally indicated), while keeping its per-phase impedance $\bar{Z}_2$ the same as computed in part c. What is the new total line current (magnitude) being supplied by the source?

$I_{L,new} =$ _____ Unit= _____

Problem 3 (25 Points)

Consider the structure below with the given dimensions. The iron core has relative permeability $\mu_r = 11$. The length of the air gap $g = 1$ cm. The number of turns is 100. The structure has a depth of 3 cm. Fringing effects can be neglected.



a) Draw the magnetic circuit for the structure. Label direction of magnetic flux in the left, right and center legs, polarity of MMF source and all reluctances. Use the magnetic circuit to compute the inductance of the structure.

L = _____ Unit= _____

b) How much current i is needed to achieve a flux density of $B = 0.5 \text{ T}$ in the center leg?

$i =$ _____ Unit= _____

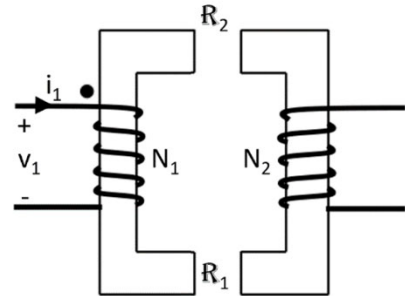
Problem 4 (25 Points)

Please show your calculations and steps to receive full credit.

Writing just the correct numerical answer with no derivation will not result in credit.

Writing just the formula will not result in partial credit.

a) Two coils are wrapped around an iron core as shown (not to scale). Coil 1 on the left has 10 turns, and Coil 2 on the right has 20 turns. The two air gaps have reluctances $R_1 = 5 \text{ At/Wb}$, $R_2 = 15 \text{ At/Wb}$. The reluctance of the core can be neglected.



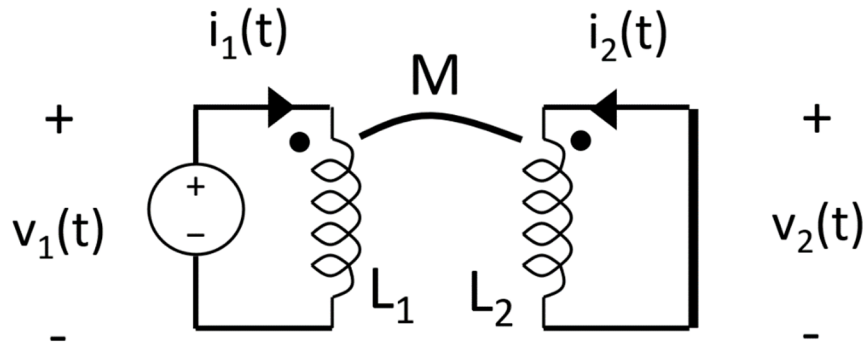
What is the self-inductance for each coil L_1 and L_2 and the mutual inductance between the two coils M ? (Recall that M should always be a positive number.)

$$L_1 = \underline{\hspace{2cm}} \text{ Unit} = \underline{\hspace{1cm}} \quad L_2 = \underline{\hspace{2cm}} \text{ Unit} = \underline{\hspace{1cm}}$$

$$M = \underline{\hspace{2cm}} \text{ Unit} = \underline{\hspace{1cm}}$$

Part b) is unrelated to part a).

b) The coupled coils below have $L_1 = 10 \text{ H}$, $L_2 = 5 \text{ H}$ and $M = 5 \text{ H}$. The primary is driven by a voltage source and the secondary has been short-circuited.



If $v_1(t) = \sqrt{2} \cos(120\pi t)$ is applied to the first coil, what is the short-circuit current $i_2(t)$ observed at the second coil?

Please give an answer as a function of time. Please ensure that your answer is consistent with the polarity of the currents as indicated in the diagram.

$i_2(t) =$ _____ Unit= _____